Interpretable Machine Learning as an Exploratory Data Analysis Tool: An Application to Post-Pricing Validation of Japanese Private Health Insurance Products

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Abstract

This paper demonstrates the usefulness of Interpretable Machine Learning (IML) as an Exploratory Data Analysis (EDA) tool for actuarial practice through its application to post-pricing validation of private health insurance products sold in Japan. Traditional actuarial practice in Japan has limitations in identifying important features, examining their feature effects, and detecting complex interactions among multiple features. To address these limitations, this study applies several IML methods, including Permutation Feature Importance (PFI), Partial Dependence Plot (PDP), Friedman's H-statistic, and Shapley additive explanations (SHAP), thereby complementing existing actuarial practice. The analysis is conducted using simulated policy data designed to mimic the structure of insurance operational data, including the data fields, aggregation framework, and data generation process, while preserving data confidentiality. In addition, practical implementation issues, including computational cost and dataset size, are discussed to facilitate the adoption of IML in actuarial practice.

Keywords: Interpretable Machine Learning, Exploratory Data Analysis, Private Health Insurance, Morbidity Rate, Actuarial Practice, Post-Pricing Validation, Actual-to-Expected Ratio

1 Introduction

This paper aims to demonstrate that Interpretable Machine Learning (IML) [1] is a useful Exploratory Data Analysis (EDA) [2] tool for actuarial practice. While Machine Learning (ML) provides highly accurate predictive models, it is often criticized for its lack of interpretability. To address this issue, IML has been developed as a collection of methods for interpreting and visualizing the predictions of machine learning models and the effects of their features. Methods such as Permutation Feature Importance (PFI) [3], Partial Dependence Plot (PDP) [4], Friedman's H-statistic [5], and Shapley additive explanations (SHAP) [6] enable practitioners to analyze feature importance, nonlinear feature effects, feature interactions, and feature contributions to individual predictions, even for black-box machine learning models. Consequently, IML is valuable not only for explaining model predictions but also as an EDA tool for exploring complex interaction effects among multiple features. This capability is particularly beneficial in actuarial practice, where numerous risk factors and policy attributes interact in complex ways. IML can therefore complement traditional aggregated analyses by revealing patterns that are difficult to identify using existing approaches.

This paper presents an application of IML to post-pricing validation of private health insurance products sold in Japan. Health insurance is particularly sensitive to advances in medical technology, changes in healthcare utilization, and revisions to public healthcare systems, making future claim experience highly uncertain. Consequently, insurers continuously compare actual

experience with pricing assumptions after product launch and consider actions such as premium rate reviews when necessary. In Japan, premiums are generally determined using morbidity rate assumptions that include safety margins, and post-pricing validation is performed by comparing actual experience with these pricing assumptions. In practice, Actual-to-Expected (A/E) ratios are calculated for segments defined by factors such as gender, age group, and duration. This traditional A/E analysis is widely used in Japanese life insurance companies to evaluate the appropriateness of pricing assumptions.

However, traditional A/E analysis in Japan has several limitations. First, the features selected for analysis often depend on analysts' experience or prior hypotheses. As a result, important features affecting A/E ratios may be overlooked. Second, because observed A/E ratios also reflect differences in the distributions of other features, it is difficult to evaluate the effect of each feature separately from these differences. Third, understanding the factors associated with high A/E ratios often requires examining interactions among multiple features. However, analyses based solely on analysts' prior knowledge may fail to identify unexpected but important feature combinations. Finally, detailed analyses involving multiple features inevitably divide the data into increasingly smaller segments, making it difficult to determine whether observed differences represent genuine patterns or random variation.

To address these limitations, this paper applies IML techniques to post-pricing validation. Specifically, PFI, PDP, and Friedman's H-statistic are used to analyze feature importance, feature effects, and feature interactions that are difficult to identify using traditional A/E analysis, thereby complementing current actuarial practice. Furthermore, policy-level analysis using SHAP visualizes the contribution of each feature to the predicted A/E ratio of individual policies, examining feature-contribution patterns for policies with high predicted A/E ratios.

This paper makes three main contributions.

First, this paper demonstrates the use of IML not for explaining a predictive model already adopted for operational use, but as an EDA tool that complements traditional A/E analysis by exploring patterns in the data and generating hypotheses for subsequent modeling and actuarial analysis. In recent years, many actuarial studies have applied machine learning to pricing in order to improve the prediction of mortality and morbidity rates [7, 8]. In these studies, IML is primarily used to interpret machine learning models. However, applying machine learning directly to pricing still faces practical challenges, including explainability requirements and regulatory considerations, limiting its adoption in practice. In contrast, the approach proposed in this paper can be incorporated into existing A/E analysis without fundamentally changing current actuarial workflows, making it a practical and accessible application of IML.

Second, this paper demonstrates the proposed framework using a synthetic dataset designed to mimic the structure of insurance operational data from Japanese life insurance companies. Real-world insurance operational data differ substantially from publicly available health insurance datasets because they contain numerous policy attributes, high-cardinality categorical features, and actuarial variables required for A/E analysis. Therefore, the synthetic dataset was constructed to reproduce these practical characteristics while protecting confidential business information. This enables the proposed framework to be demonstrated under data conditions that closely resemble those encountered in actuarial practice.

Finally, to facilitate practical adoption, the Python code and simulated dataset used in this study will be made publicly available through GitHub, allowing readers to apply the proposed framework to their own actuarial analyses. In addition, this paper discusses practical implementation considerations, including model development and data preprocessing, for practitioners who are less familiar with machine learning. The effects of dataset size on IML analyses and the computational costs of the methods are also examined to support practical implementation.

The remainder of this paper is organized as follows. Section 2 introduces the IML methods used in this study. Section 3 reviews the traditional A/E analysis currently used in Japanese life insurance practice and discusses its limitations. Section 4 describes the analytical datasets

and the predictive modeling framework. Section 5 presents the results of the IML analyses and compares them with those obtained from traditional A/E analysis. Section 6 discusses the practical implications, limitations and future directions of the proposed framework. Section 7 concludes the paper.

2 Interpretable Machine Learning

2.1 Overview

IML refers to a collection of methods for interpreting and visualizing the predictions of machine learning models and the effects of their features. While machine learning models, such as gradient boosting and neural networks, often achieve high predictive performance, they are frequently criticized for their lack of interpretability. In many practical applications, it is important not only to obtain accurate predictions but also to understand and explain the relationships between input features and model predictions. Consequently, IML has attracted increasing attention as a means of interpreting the predictions and behavior of machine learning models.

IML is commonly used to explain model predictions. In this paper, however, IML is employed as a tool for EDA. EDA refers to the process of exploring data to identify influential features and understand feature effects and interaction effects. By fitting a flexible machine learning model to the data and examining its prediction behavior using IML, analysts can explore patterns represented by the fitted model, including feature effects and interactions. Based on this idea, this paper uses IML as a tool for EDA.

IML methods can be broadly categorized from two perspectives. One distinction is between model-specific methods, which are designed for particular machine learning models, and model-agnostic methods, which can be applied regardless of model type. For example, the regression coefficients of a linear regression model provide a model-specific interpretation. Another distinction is between global interpretability, which explains the overall behavior of a model, and local interpretability, which explains predictions on each individual record.

This paper employs four representative IML methods: PFI, PDP, Friedman's H-statistic, and SHAP. PFI, PDP, and Friedman's H-statistic are model-agnostic methods, whereas SHAP is implemented using TreeSHAP for the tree-based prediction model adopted in this study. PFI, PDP, and SHAP were selected because they are widely used and supported by commonly used software packages. Friedman's H-statistic was selected to quantify the strength of pairwise interaction effects together with two-way PDP. The choice of these methods does not imply that they are always the most appropriate. Other IML methods may be more suitable depending on the purpose of the analysis and the characteristics of the data. The following sections describe the purpose, basic methodology, and practical considerations of each method. Examples of the analytical results obtained using these methods are presented in Section 5 using the simulated dataset.

2.2 Permutation Feature Importance

PFI is a method for quantifying the importance of each feature in a predictive model.

The concept of permutation feature importance was originally introduced by Breiman for Random Forests [9]. Building on this idea, Fisher et al. proposed a model-agnostic measure of feature importance [3]. In this paper, PFI refers to the model-agnostic feature importance proposed by Fisher et al.

PFI measures feature importance by evaluating the increase in prediction error after randomly permuting the values of a feature. PFI is generally defined as

$$PFI_j = L(f(X_{perm(j)}), y) - L(f(X), y)$$

where $f(\cdot)$ denotes the trained model, $L(\cdot)$ denotes the prediction error measure used to evaluate model performance, X denotes the observed feature values, y denotes the corresponding target values, and $X_{perm(j)}$ denotes the feature values obtained by randomly permuting only feature j. Thus, PFI measures the deterioration in predictive performance caused by randomly permuting feature j. A larger PFI value indicates that the feature is more important for model prediction. In practice, the permutation is often repeated several times, and the resulting increases in prediction error are averaged to reduce variability due to random permutation.

PFI is classified as a global interpretability method because it evaluates feature importance for the model as a whole.

An important advantage of PFI is that it quantifies the importance of all features on a common scale, enabling their relative importance for model prediction to be compared under a common evaluation measure.

On the other hand, when features are highly correlated, permuting a feature may create unrealistic combinations of feature values that have little or no support under the joint distribution of the features, resulting in misleading importance estimates. Furthermore, when multiple correlated features contain similar information, the model may use them interchangeably. As a result, their importance may be distributed across those features, causing the importance of genuinely influential features to be underestimated. Therefore, when evaluating feature importance using PFI, it is desirable to examine the correlation structure among features as well. When the features are strongly correlated, other methods for evaluating feature importance may also be considered.

2.3 Partial Dependence Plot

PDP is a method for visualizing how the predicted value changes on average as the value of a feature changes. It was proposed by Friedman in 2001 [4].

The partial dependence (PD) function estimates the average predicted value by fixing the value of the feature of interest while averaging over the values of all other features. Visualizing this function as a PDP enables the relationship between the feature and the model prediction to be examined after averaging over the values of the other features. In particular, the PDP can reveal whether the predicted value increases or decreases as the feature value changes, as well as whether the relationship is linear, exhibits a threshold effect, or follows a nonlinear pattern such as saturation.

For a single feature, the corresponding partial dependence function, referred to as the one-way PD, is generally defined as

$$PD_j(x_j) = E_{X_{-j}}[f(x_j, X_{-j})]$$

where $f(\cdot)$ denotes the trained model, x_j is the value of feature j, and X_{-j} represents all remaining features. The visualization of this function is referred to as the one-way PDP.

PDP can also be extended to two features. For two features, the corresponding partial dependence function, referred to as the two-way PD, is generally defined as

$$PD_{jk}(x_j, x_k) = E_{X_{-(j,k)}}[f(x_j, x_k, X_{-(j,k)})]$$

where x_j and x_k are the values of the two features of interest, and $X_{-(j,k)}$ represents all remaining features. The visualization of this function is referred to as the two-way PDP.

Similar to PFI, PDPs may produce misleading results when features are strongly correlated. Because the PD function is computed by changing only the feature of interest while keeping all other feature values unchanged, it may generate combinations of feature values that have little or no support under the joint distribution of the features when features are strongly correlated. For example, if only one of two highly correlated features is modified, the predicted value may be evaluated for an unrealistic combination of feature values. Therefore, when examining feature

effects using a PDP, it is important to consider the correlation structure among features. When features are strongly correlated, alternative methods for examining feature effects may also be considered.

2.4 Friedman's H-statistic

Friedman's H-statistic is a method for quantifying the strength of interaction effects between features. It was proposed by Friedman and Popescu in 2008 [5].

In predictive models, predictions may depend not only on the additive effects of individual features but also on interactions among multiple features. For example, if the effect of feature j on the prediction varies depending on the value of another feature k, an interaction is considered to exist between the two features.

The H-statistic extracts the interaction component by removing the main effects of the individual features from the two-way PD function and quantifies its magnitude. Before calculating the H-statistic, the one-way and two-way PD functions are centered so that their mean over the observations is zero. Let $\widetilde{PD}_j(x_j)$, $\widetilde{PD}_k(x_k)$, and $\widetilde{PD}_{jk}(x_j, x_k)$ denote these centered partial dependence functions. Then, for two features j and k, the H-statistic can be expressed as

$$H_{jk}^2 = \frac{\sum_{i=1}^{n} [\widetilde{PD}_{jk}(x_j^{(i)}, x_k^{(i)}) - \widetilde{PD}_j(x_j^{(i)}) - \widetilde{PD}_k(x_k^{(i)})]^2}{\sum_{i=1}^{n} \widetilde{PD}_{jk}^2(x_j^{(i)}, x_k^{(i)})}.$$

where n denotes the number of observations.

The numerator represents the variance of the interaction component obtained by removing the main effects from the two-way PD function, whereas the denominator represents the variance of the two-way PD function. Thus, the H-statistic quantifies the strength of interaction relative to the overall variation in the two-way PD function.

The statistic represents a proportion and therefore is expected to take values between 0 and 1, although the computed value may occasionally exceed 1 in practice. A value close to 0 indicates little or no interaction, whereas a value close to 1 indicates a strong interaction between the features.

Because the H-statistic is calculated using PD functions, it is also affected by the limitations of PD when features are strongly correlated. In addition, when only a small number of observations are available in certain regions of the feature space, the estimated interaction effects may become unstable. Therefore, when using the H-statistic to evaluate interaction effects, it is important to consider the data distribution and the correlation structure among features.

2.5 SHAP

SHAP is a method for explaining the contribution of individual features to a single prediction. It was proposed by Lundberg and Lee in 2017 [6]. SHAP is based on the Shapley value from cooperative game theory.

The concept of the Shapley value can be illustrated using a simple example. Suppose that three players, A, B, and C, cooperate in a game, and that the reward depends on the combination of participating players, as shown in Table 1.

As shown in Table 2, the contribution of each player depends on the order in which the players join the game.

With three players, there are 3!=6 possible orders in which the players can join the game. The Shapley value is defined as the average marginal contribution of each player over all possible orders. In this example, the Shapley value of player A is 38.3.

SHAP applies the concept of the Shapley value to the interpretation of predictive models. Each feature is regarded as a player in the game, and a game reward is assigned to each set of

Table 1: Rewards for each combination of players

Participants	Reward
None	0
A	20
B	15
C	10
A, B	55
A, C	50
B, C	45
A, B, C	100

Table 2: Marginal contribution of each player under different participation orders

Order	A	B	C
A $\rightarrow$ B $\rightarrow$ C	20	35	45
A $\rightarrow$ C $\rightarrow$ B	20	50	30
B $\rightarrow$ A $\rightarrow$ C	40	15	45
B $\rightarrow$ C $\rightarrow$ A	55	15	30
C $\rightarrow$ A $\rightarrow$ B	40	50	10
C $\rightarrow$ B $\rightarrow$ A	55	35	10
Average	38.3	33.3	28.3

players based on the model and the corresponding set of features. The definition and computation of this reward depend on the SHAP formulation and implementation. The contribution of a feature is defined as the change in this value when the feature is added to the set. The SHAP value of each feature is then obtained by averaging this contribution over all possible orders in which the features are added.

The model prediction can be expressed as the sum of the baseline prediction and SHAP values of all features:

$$f(x) = \phi_0 + \sum_{j=1}^{M} \phi_j,$$

where $f(x)$ is the prediction for the observation of interest, ϕ_0 denotes the baseline model output used in the chosen SHAP implementation, ϕ_j is the SHAP value of feature j, and M is the number of features.

A positive SHAP value indicates that the feature increases the prediction, whereas a negative SHAP value indicates that it decreases the prediction. The larger the absolute SHAP value, the greater the contribution of the feature to the prediction.

SHAP itself is a local interpretability method because it explains the prediction for an individual observation. A commonly used visualization for local interpretation is the SHAP waterfall plot, which shows how the prediction is obtained by starting from the baseline prediction and sequentially adding the contributions of individual features. It enables users to identify which features increase or decrease the prediction for a particular observation.

SHAP can also be used for global interpretability. A commonly used visualization for this purpose is the SHAP summary plot, which displays the SHAP values of all observations simultaneously. This makes it possible to examine feature importance and the distribution of feature contributions. Thus, SHAP can be used not only to explain individual predictions but also to analyze the overall behavior of a predictive model.

However, SHAP values depend on the rule used to assign a game reward to each set of features. They may also depend on the background data and the assumptions made about

dependence among features. Therefore, different definitions or implementations may produce different SHAP values. These points should be considered when interpreting SHAP values [10, 11].

The calculation of SHAP values requires assigning a value to each subset of features according to a specified rule for handling features not included in that subset. A direct calculation from the definition of the Shapley value involves all possible subsets of features (2^M subsets). Consequently, the computational cost increases rapidly as the number of features grows, making exact computation impractical for models with many features.

For decision tree models, an efficient algorithm called TreeSHAP has been proposed [12]. TreeSHAP uses the structure of decision trees to compute SHAP values efficiently.

3 Traditional A/E Analysis in Japan

3.1 Overview

This paper presents an application of IML to post-pricing validation for private health insurance in Japan. This section first provides an overview of the traditional A/E analysis used in Japan. It then describes the practical workflow and limitations of traditional A/E analysis. Finally, it introduces the lump-sum hospitalization rider, which serves as the case study in this paper.

In Japan, premiums for private health insurance products are generally determined using morbidity rate assumptions that include safety margins to account for future uncertainty and risk. After a product is launched, insurers continuously monitor the difference between actual experience and the morbidity rate assumptions used in pricing. Specifically, for each segment defined by factors such as gender, age group, and duration, the Actual-to-Expected (A/E) ratio is calculated by dividing the actual claim amount by the expected claim amount based on the pricing assumptions.

$$\text{A/E ratio} = \frac{\sum \text{Actual Claim Amount}}{\sum \text{Expected Claim Amount}}$$

Although the concept of A/E analysis is widely used internationally for various purposes, such as validating valuation assumptions, the A/E analysis considered in this paper specifically refers to post-pricing validation, which assesses whether actual claim experience remains within the range anticipated by the morbidity rate assumptions used in pricing.

Traditional A/E analysis in Japan offers several practical advantages. First, because the A/E ratio is calculated based on claim amounts, it directly reflects the financial impact on the insurer. Second, the calculation is straightforward, and an A/E ratio exceeding 100% indicates that actual experience is worse than expected, making the results relatively easy to explain even to non-actuaries. Third, A/E analysis is widely used for regulatory monitoring and other actuarial purposes, and most Japanese life insurance companies have already established operational frameworks for data aggregation and calculation. Consequently, traditional A/E analysis in Japan continues to play an important role in private health insurance practice in Japan.

3.2 Current Practice

In practice, insurers perform various types of A/E analyses, including single-year analysis, year-over-year comparison, historical trend analysis, and future projections. This paper focuses on the most fundamental approach, namely single-year analysis.

Japanese life insurance companies generally maintain policy-level expected and actual claim amounts in their information systems based on the morbidity assumptions, policy benefits, and policy information used in pricing. Table 3 presents an example of the policy-level data managed by the system.

Analysts specify aggregation keys such as product, gender, attained age, and policy duration to obtain A/E ratios for individual segments. Table 4 shows an example of aggregated results

Table 3: Example of policy-level data maintained in the operational system

Policy ID	Gender	Attained Age	Policy Duration	Sum Assured	Morbidity Rate	Expected Claim Amount	Actual Claim Amount
P0001	Male	30	2	50,000	0.0018	90	0
P0002	Female	45	6	100,000	0.0035	350	100,000
P0003	Male	57	12	200,000	0.0050	1,000	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Table 4: Example of A/E analysis results aggregated by gender and attained age

Gender	Attained Age	Expected Claim Amount	Actual Claim Amount	A/E Ratio
Male	30–34	9,500	7,800	82.1%
Male	35–39	12,300	13,600	110.6%
Male	40–44	18,100	17,900	98.9%
⋮	⋮	⋮	⋮	⋮
Female	30–34	8,900	8,500	95.5%
⋮	⋮	⋮	⋮	⋮
Total	–	1,520,000	1,476,000	97.1%

by gender and attained age. These aggregated A/E ratios are then used to examine the level and overall pattern of claim experience across different segments.

Figure 1 shows the A/E ratios by gender and attained age. Because morbidity assumptions used in pricing are often established separately by gender and attained age, this analysis enables actuaries to identify directly which age groups or genders deviate most from the pricing assumptions. In practice, attained ages are typically grouped appropriately to avoid excessively small segments, and analysts visually examine the overall shape of the A/E curve as well as segments with particularly high A/E ratios.

Figure 2 presents A/E ratios by policy duration. Claim incidence is generally lower during the early policy years because of selection effects and gradually increases before reaching a stable level. Since pricing assumptions are generally based on the morbidity rate after the selection effect has disappeared, this analysis enables actuaries to assess the adequacy of the pricing assumptions without being affected by the temporary selection effect.

Analysis by gender and attained age and analysis by policy duration are regularly performed as part of A/E analysis. In addition to these standard analyses, further investigations may be conducted based on the analysts' experience or prior hypotheses. Figure 3 presents an example of A/E ratios by sum assured. If policies with larger sum assured exhibit higher A/E ratios, this may indicate potential moral hazard.

When more detailed investigation is required, analysts may construct a policy-level analytical dataset from policy-level records rather than relying solely on the aggregated data maintained in the operational system. This analytical dataset can incorporate not only policy information stored in the operational database but also additional variables created for analytical purposes or obtained from other internal data sources, such as rider information and underwriting information.

Thus, in practice, insurers first identify overall trends using aggregated A/E analysis and, when necessary, prepare a policy-level analytical dataset for further investigation. This staged analytical process is widely adopted in Japanese private health insurance practice.

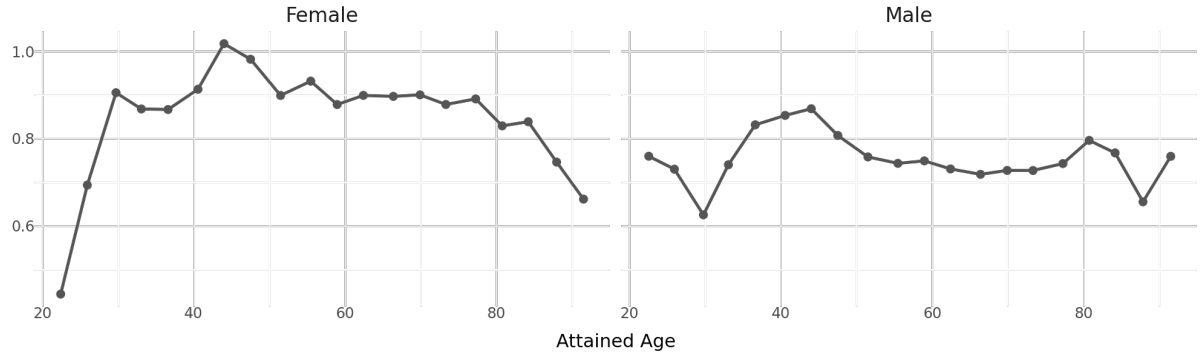


Figure 1: A/E ratios by gender and attained age

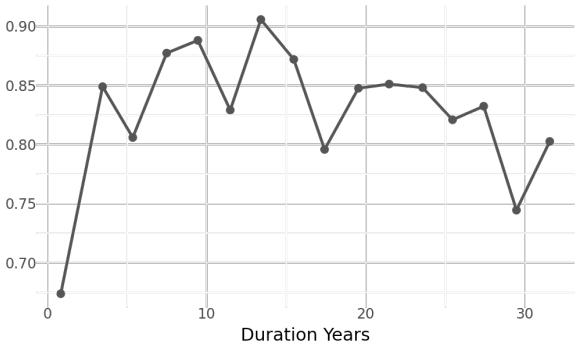


Figure 2: A/E ratios by duration

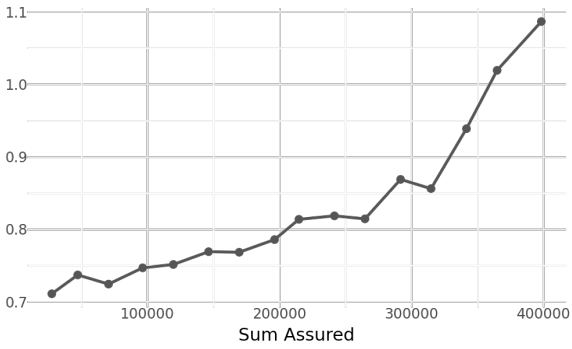


Figure 3: A/E ratios by sum assured

3.3 Challenges of Current Practice

The current aggregation-based A/E analysis has several limitations.

First, the selection of features for analysis largely depends on established practice, the analyst's experience or prior hypotheses. In practice, insurers regularly conduct standard analyses, such as A/E ratios by gender and attained age or by policy duration, and subsequently perform additional analyses for features that analysts consider relevant based on the initial results. However, real-world insurance data contain a large number of features, including policy attributes, product characteristics, and underwriting information. Examining A/E ratios for every available feature is therefore impractical. As a result, even if a feature has a substantial impact on the A/E ratio, it may not be analyzed unless it is included in the analyst's initial hypotheses, potentially leading to important insights being overlooked.

Second, it is difficult to evaluate the effects of individual features separately from differences in portfolio composition. For example, Figure 1 shows relatively high A/E ratios for lives insured aged 30–49. However, this observation alone does not indicate whether the higher A/E ratios are attributable to age itself or to differences in other policy characteristics. For example, if lives insured aged 30–49 are more likely to purchase policies with larger sums assured, and such policies tend to exhibit higher A/E ratios, then the observed age-specific A/E ratios also reflect differences in the distribution of the sum assured. In other words, even A/E ratios aggregated by age may incorporate the effects of features other than age. As demonstrated later in this paper, the IML analysis reveals that much of the observed age-related difference is actually explained by differences in the composition of policy characteristics. Thus, aggregation-based analysis generally reflects both the feature of interest and differences in the composition of other features, making it difficult to evaluate the effect of an individual feature separately from differences in the composition of the other features.

Third, identifying interaction effects among features is challenging. In practice, when a

segment exhibits a particularly high A/E ratio, analysts investigate the underlying factors to determine whether premium rates or product design should be reviewed. However, elevated A/E ratios cannot always be explained by the effects of individual features alone. Instead, they may become apparent only through specific combinations of features, such as lives insured with longer policy durations and larger sums assured. Understanding such interaction effects is essential for identifying the underlying drivers of high A/E ratios. Nevertheless, current aggregation-based analyses primarily examine each feature separately, such as attained age or policy duration, making it difficult to identify systematically which combinations of features should be investigated. Although analysts can examine selected feature combinations based on their experience or prior hypotheses, important interactions may remain undetected if they involve combinations that were not considered in advance.

Fourth, it is difficult to investigate high A/E ratios in detail from a multivariable perspective. In practice, analysts often wish to examine multiple features simultaneously, such as age, policy duration, and sum assured, in order to understand the factors associated with high A/E ratios. However, under an aggregation-based approach, the only way to analyze multiple features simultaneously is to subdivide the data into increasingly detailed segments defined by combinations of these features. As a result, both the number of policies and the expected claim amount within each segment decrease substantially. For example, a segment defined by "female lives insured aged 30–49 with a policy duration of three years and a large sum assured" may contain only a small number of policies. Consequently, the A/E ratio of such a segment may fluctuate considerably because of only a few claims, making it difficult to distinguish genuine patterns from random variation. Therefore, although detailed multivariable analyses are desirable in practice, aggregation-based analyses are constrained by data sparsity caused by excessive segmentation.

3.4 Lump-Sum Hospitalization Rider

The lump-sum hospitalization rider is a common rider to private health insurance policies sold by Japanese life insurance companies. It pays a fixed benefit when the life insured is hospitalized because of illness or injury. It is one of the most widely marketed health insurance riders in the Japanese insurance market. Traditionally, Japanese health insurance products primarily provided daily hospitalization benefits, under which benefit payments were proportional to the length of hospitalization. In recent years, however, advances in medical technology and changes in clinical practice have shortened the average length of hospital stays, increasing demand for lump-sum hospitalization benefits that provide a fixed amount of coverage even for short-term hospitalization. Under this rider, benefits may be paid repeatedly for the same policy. The sum assured can be selected by the policyholder within a predetermined range.

The lump-sum hospitalization rider is also an important target of regular monitoring in actuarial practice. Unlike daily hospitalization benefits, whose payment amount depends on the length of hospitalization, the lump-sum benefit pays a fixed amount regardless of the duration of the hospital stay. Consequently, relatively large benefit payments may be made even for minor hospitalizations. If policies with larger sums assured exhibit higher A/E ratios, this may indicate potential moral hazard on the one hand, such as purchasing insurance or submitting claims primarily to obtain benefit payments. On the other hand, it is generally difficult to determine objectively whether an individual claim is motivated by moral hazard, and many such claims are paid as legitimate insurance claims. Therefore, in practice, insurers analyze A/E ratios to identify the policy characteristics associated with segments where potential moral hazard may be more prevalent and, when necessary, consider measures such as revising underwriting guidelines or strengthening ongoing monitoring.

4 Data and Modeling Framework

4.1 Overview of Analytical Cases

Based on the practical workflow described in the previous section, this study considers two analytical cases to demonstrate how IML can be applied to A/E ratio analysis. The analysis focuses on the lump-sum hospitalization rider. Only standard-risk policies are included, while policies issued with special underwriting conditions, such as medical loadings or policy exclusions, are excluded from the analysis.

The data used in this study are synthetic data constructed with reference to actual operational data from a Japanese life insurance company. To protect confidentiality, the dataset does not reproduce the actual portfolio characteristics or the interaction effects among features. Instead, while preserving the structure of insurance operational data, it was intentionally designed so that the characteristics of each IML method could be clearly illustrated. Therefore, the objective of this study is not to quantify the effects of individual features in actual health insurance portfolios, but rather to demonstrate how IML can provide insights that are difficult to obtain through traditional aggregation-based A/E analysis.

In Case 1, the analysis is conducted using Dataset A, which consists of aggregated data extracted from the A/E analysis database. Although, in practice, analysts often examine only a subset of features considered important based on experience, this study analyzes all features available in the database. The objective of Case 1 is to obtain a global understanding of the features influential in the fitted model and the interaction effects among those features.

In Case 2, the analysis is conducted using Dataset B, which contains policy-level data. Compared with Dataset A, Dataset B includes more granular policy and product attributes, enabling more comprehensive analyses at the individual policy level. The objective of Case 2 is to understand the factors associated with policies exhibiting high predicted A/E ratios and to examine how each feature contributes to the predicted A/E ratio for individual policies.

An overview of Cases 1 and 2 is presented in Table 5.

Table 5: Overview of analytical cases

Case	Purpose	Dataset	IML Methods
Case 1	Obtain a global understanding of the features influential in the fitted model and their interaction effects	Dataset A	PFI, SHAP summary plot, PDP (including two-way PDP), Friedman's H-statistic
Case 2	Understand the factors associated with policies exhibiting high predicted A/E ratios at the individual policy level	Dataset B	SHAP waterfall plot

4.2 Datasets

4.2.1 Dataset A: Aggregate Data

Table 6 summarizes the variables included in Dataset A. The table presents the variables in Dataset A, including the features used as model inputs and the variables used to define the response and sample weights.

In operational A/E analysis databases, categorical variables often contain a large number of categories, while numerical variables retain detailed policy-level values. In this study, however, several variables were preprocessed by combining categories or converting continuous values into grouped categories before model development, following common actuarial practice. Examples of the original values in operational data and the grouping schemes adopted in this study are provided in Appendix A.

The purpose of combining categorical variables was not to improve predictive performance but to facilitate exploratory data analysis using IML. Variables with many categories often contain only a small number of observations within individual categories, increasing the risk that the model overfits category-specific patterns. Furthermore, visualizations such as PDPs and SHAP summary plots become difficult to interpret when they contain a large number of categories, making it difficult to identify the overall trends.

Similarly, numerical variables were grouped according to the aggregation levels commonly used in actuarial practice. For example, attained age was grouped into five-year bands, and the sum assured was grouped into intervals of JPY 100,000. The purpose of this preprocessing was not to improve predictive accuracy but to present the analytical results at a level of granularity that is familiar to practitioners and therefore easier to interpret.

Table 6: Variables included in Dataset A

Variable	Description	Type	Possible Values
`observation_year`	Observation year	Integer	2025
`rider_product`	Rider product code (analysis restricted to one rider product)	Categorical	05 (= Lump-Sum Hospitalization Rider)
`gender`	Gender of the life insured	Categorical	Male, Female
`attained_age_band`	Attained age of the life insured	Categorical	15–19, 20–24, . . . , 100–104
`duration_year_band`	Policy duration	Categorical	0, 1, 2, . . . , 14, 15+
`conversion_indicator`	Whether the policy was issued as new business or through conversion[1]	Categorical	Conversion, New Business
`rider_attachment_timing`	Timing of rider attachment	Categorical	At-Issue, After-Issue
`renewal_status`	Whether the rider is before or after the first renewal	Categorical	Pre-Renewal, Post-Renewal
`sum_assured_band`	Sum assured group (unit: JPY 10,000)	Categorical	1–10, 11–20, 21–30, 31–40
`base_product_category`	Base product category	Categorical	Medical, Renewable Term, Term, Cancer, Whole Life
`underwriting_method`	Underwriting method	Categorical	Declaration, Health Check-up, Medical Examination
`distribution_channel`	Distribution channel	Categorical	Career Agent, Bank Agency, Independent Agency, Direct Marketing
`sales_prefecture_group`	Region where the policy was sold	Categorical	Hokkaido, Tohoku, Kanto, Chubu, Kinki, Chugoku, Shikoku, Kyushu
`corporate_indicator`	Whether the policyholder is an individual or a corporation	Categorical	Yes, No
`actual_claim_amount`	Total actual claim amount during the observation year	Integer	≥ 0
`expected_claim_amount`	Total expected claim amount based on pricing assumptions during the observation year	Float	> 0

4.2.2 Dataset B: Policy-Level Data

Compared with Dataset A, Dataset B incorporates additional policy- and product-level attributes that are not captured in Dataset A. Specifically, it contains information on the sums assured of other riders, such as the cancer benefit rider, nursing care benefit rider, dementia benefit rider, and female-specific disease benefit rider. It also includes indicators showing whether

[1]Conversion refers to replacing an existing life insurance policy with a new policy, with the value of the existing policy being applied toward the new policy. It is a common practice in the Japanese life insurance market.

optional riders, such as the premium waiver rider and the advanced medical treatment rider, are attached to the policy.

The variables included in Dataset B are summarized in Table 7.

Table 7: Variables included in Dataset B

Variable	Description	Type	Possible Values
observation_year	Observation year	Integer	2025
policy_id	Policy identifier	Categorical	01-013878, ...
issue_date	Issue date	Integer	20241018, ...
gender	Gender of the life insured	Categorical	Male, Female
attained_age	Attained age of the life insured	Integer	15, 16, ..., 103, 104
duration_year	Policy duration	Integer	0, 1, 2, ..., 32, 33
conversion_indicator	Whether the policy was issued as new business or through conversion	Categorical	Conversion, New Business
rider_attachment_timing	Timing of rider attachment	Categorical	At-Issue, After-Issue
renewal_status	Whether the rider is before or after the first renewal	Categorical	Pre-Renewal, Post-Renewal
base_product	Base product name	Categorical	Term1992, Term2001, ..., Renewal-Term1998, ..., WholeLife1994, ... Medical2000, ..., Cancer1997,...
underwriting_method	Underwriting method	Categorical	Declaration, Health Check-up, Medical Examination
distribution_channel	Distribution channel	Categorical	Career Agent, Bank Agency, Independent Agency, Direct Marketing
sales_prefecture_group	Region where the policy was sold	Categorical	Hokkaido, Tohoku, Kanto, Chubu, Kinki, Chugoku, Shikoku, Kyushu
corporate_indicator	Whether the policyholder is an individual or a corporation	Categorical	Yes, No
sum_assured	Sum assured of the lump-sum hospitalization rider	Integer	> 0
sum_assured_medical_comp_benefit_rider	Sum assured of the mandatory medical rider	Integer	≥ 0
sum_assured_cancer_rider	Sum assured of the cancer rider	Integer	≥ 0
sum_assured_nursing_care_rider	Sum assured of the long-term care rider	Integer	≥ 0
sum_assured_dementia_rider	Sum assured of the dementia rider	Integer	≥ 0
sum_assured_female_rider	Sum assured of the female-specific disease rider	Integer	≥ 0
sum_assured_injury_rider	Sum assured of the injury rider	Integer	≥ 0
premium_waiver_rider	Whether a premium waiver rider is attached	Categorical	Yes, No
advanced_medical_rider	Whether an advanced medical treatment rider is attached	Categorical	Yes, No
number_of_riders	Number of attached riders	Integer	≥ 0
actual_claim_amount	Total actual claim amount during the observation year	Integer	≥ 0
expected_claim_amount	Total expected claim amount based on pricing assumptions during the observation year	Float	> 0
exposure	Exposure of the rider in the observation year	Float	$0 < x \leq 1$

4.3 Prediction Model

To apply IML methods, we trained LightGBM models to predict the A/E ratio. For each record i, let Actual_i and Expected_i denote the actual and expected claim amounts, respectively, and define

$$r_i = \frac{\text{Actual}_i}{\text{Expected}_i}.$$

We considered two loss functions. The first was the expected-amount-weighted squared-error loss:

$$L_{\mathrm{WLS}}(f) = \sum_{i \in \mathrm{Training}} \mathrm{Expected}_i (r_i - f(x_i))^2.$$

The second was the Poisson deviance loss:

$$L_{\mathrm{Poisson}}(f) = \sum_{i \in \mathrm{Training}} \mathrm{Expected}_i D_{\mathrm{P}}(r_i, f(x_i)),$$

where

$$D_{\mathrm{P}}(y, \mu) = 2\{y \log(y/\mu) - (y - \mu)\},$$

with the usual convention for $y = 0$. This weighted formulation is equivalent, at the level of the objective function, to applying the Poisson deviance to the actual claim amount with mean $\mathrm{Expected}_i f(x_i)$. If an independent constant prediction c_G is assigned to each segment G, both loss functions are minimized at

$$c_G = \frac{\sum_{i \in G} \mathrm{Actual}_i}{\sum_{i \in G} \mathrm{Expected}_i},$$

which is the traditional segment-level A/E ratio. Thus, the LightGBM models extend traditional A/E analysis by allowing the predicted A/E ratio to vary flexibly across multiple features and their interactions. In this study, the model trained using the weighted squared-error loss was used as the primary model for the IML analyses. Because this model directly predicts the A/E ratio, TreeSHAP provides an additive decomposition on the A/E-ratio scale.

5 Results

This section presents the results of the analyses for Case 1 and Case 2. The objective is to demonstrate how IML provides insights that are difficult to obtain through traditional aggregation-based A/E analysis and to illustrate its potential applications in actuarial practice.

5.1 Case 1: Understanding Drivers of the A/E Ratio

In Case 1, Dataset A was used to analyze the features influential in the fitted model and the interaction effects among these features. The following sections present the insights obtained using IML and compare them with those available from traditional A/E analysis.

5.1.1 Feature Importance Analysis

The results of the permutation feature importance (PFI) analysis are shown in Figure 4.

In current actuarial practice, the features included in A/E analysis are often selected based on the analyst's experience or prior hypotheses. Consequently, important features may be overlooked, and it is difficult to objectively compare the relative importance of multiple features.

The PFI analysis ranked sum assured, duration year, gender and base product as the most influential features in the fitted model. Distribution channel and attained age showed moderate importance, while the remaining features showed relatively low importance.

The results confirmed that gender, duration year, attained age and sum assured, which are regularly examined in current actuarial practice, were all relatively influential features in the fitted model. On the other hand, base product and distribution channel exhibited relatively high importance despite not being regularly emphasized in traditional A/E analysis. This suggests that traditional analysis may overlook features that are important for the fitted model's predictions.

These results demonstrate that PFI provides a systematic and quantitative means of comparing feature importance in the fitted model without relying solely on the analyst's experience or prior hypotheses.

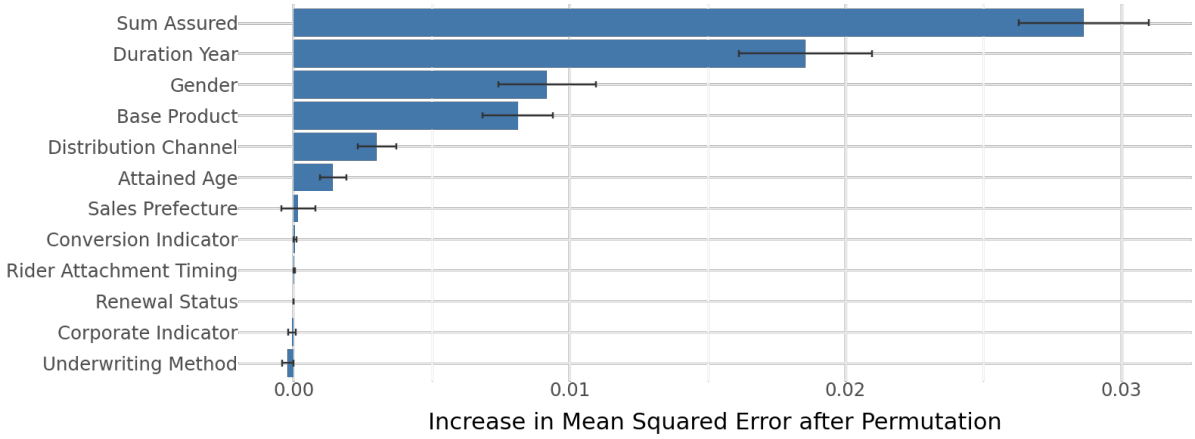


Figure 4: Key drivers of the A/E ratio identified by PFI

5.1.2 SHAP Summary Plot

Figure 5 presents the SHAP summary plot.

Unlike PFI, which summarizes feature importance as a single value, the SHAP summary plot illustrates not only the magnitude of each feature's contribution to the prediction but also the distribution and direction of these contributions across the aggregated records in Dataset A. As a result, it enables the variability and direction of feature contributions to be examined in addition to their overall importance.

The combination of PFI and the SHAP summary plot provides a more comprehensive understanding of how features contribute to the fitted model's predictions.

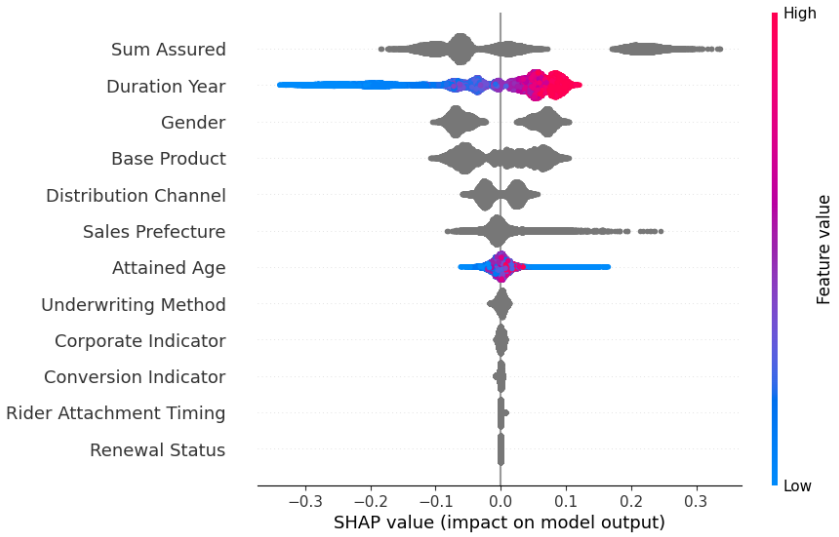
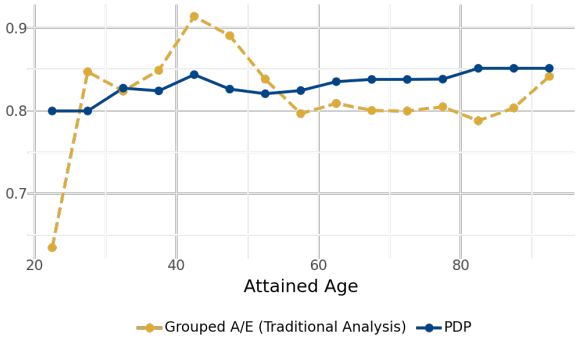
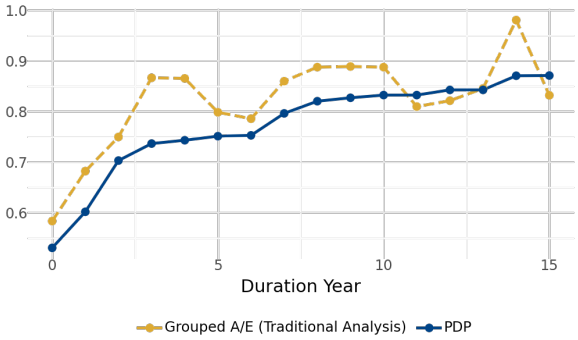


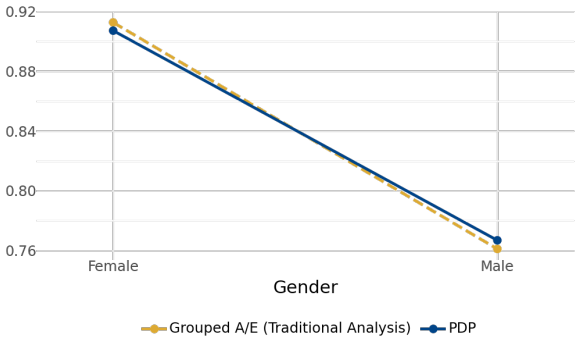
Figure 5: Distribution and direction of feature contributions shown by SHAP summary plot

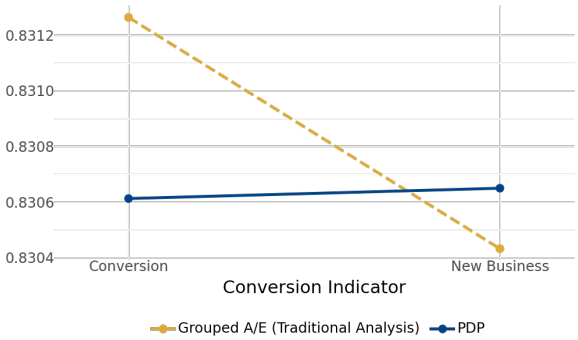
5.1.3 Partial Dependence Analysis

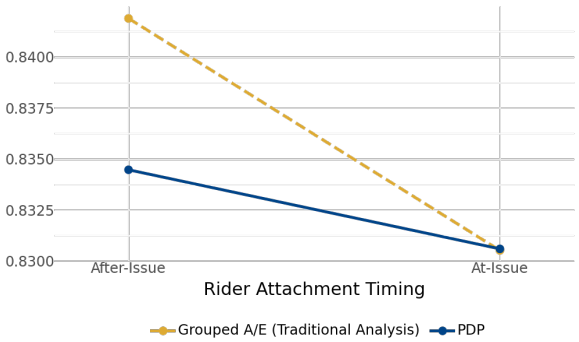
PDPs for all features are shown in Figure 6. For comparison, the corresponding results obtained from the traditional aggregation-based A/E analysis are also presented. For the PDPs shown in Figure 6, the expectation in the PD function was evaluated using the empirical distribution of Dataset A weighted by the expected claim amount.

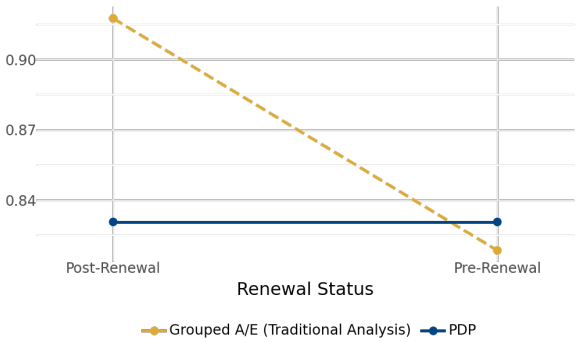

0.9
0.8
0.7
20
40
60
80
Attained Age
Grouped A/E (Traditional Analysis)
PDP

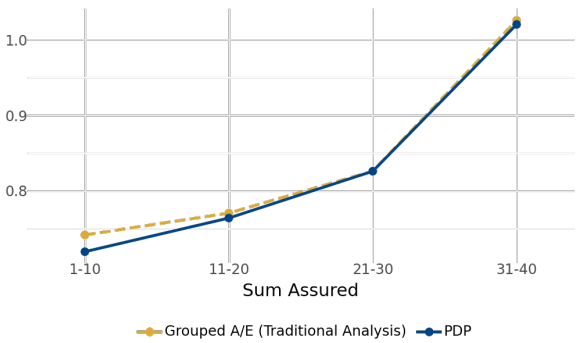

1.0
0.9
0.8
0.7
0.6
0
5
10
15
Duration Year
Grouped A/E (Traditional Analysis)
PDP

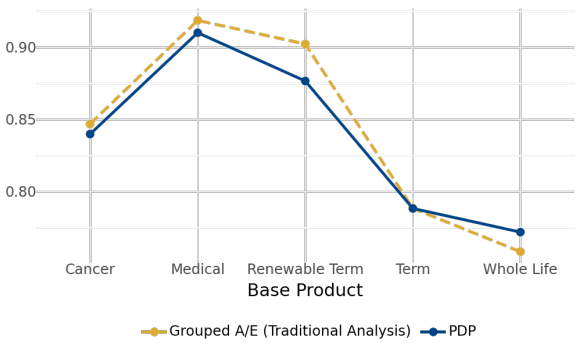

0.92
0.88
0.84
0.80
0.76
Female
Male
Gender
Grouped A/E (Traditional Analysis)
PDP

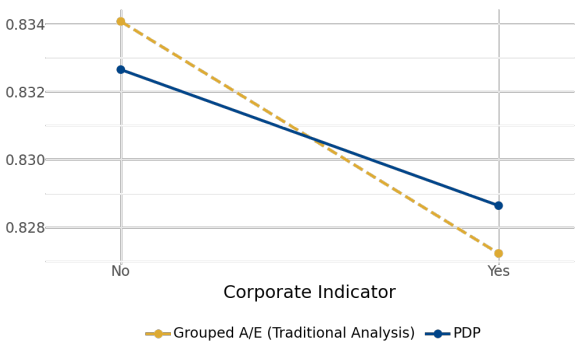

0.8312
0.8310
0.8308
0.8306
0.8304
Conversion
New Business
Conversion Indicator
Grouped A/E (Traditional Analysis)
PDP

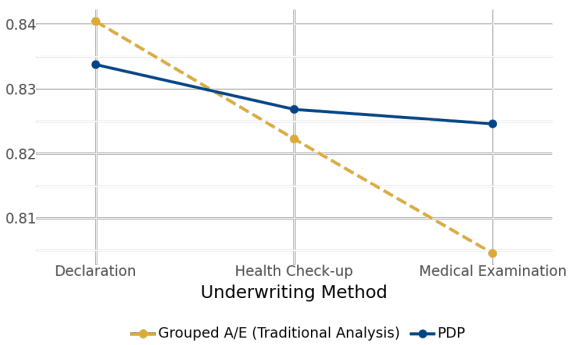

0.8400
0.8375
0.8350
0.8325
0.8300
After-Issue
At-Issue
Rider Attachment Timing
Grouped A/E (Traditional Analysis)
PDP


0.90
0.87
0.84
Post-Renewal
Pre-Renewal
Renewal Status
Grouped A/E (Traditional Analysis)
PDP


1.0
0.9
0.8
1-10
11-20
21-30
31-40
Sum Assured
Grouped A/E (Traditional Analysis)
PDP


0.90
0.85
0.80
Cancer
Medical
Renewable Term
Term
Whole Life
Base Product
Grouped A/E (Traditional Analysis)
PDP


0.834
0.832
0.830
0.828
No
Yes
Corporate Indicator
Grouped A/E (Traditional Analysis)
PDP


0.84
0.83
0.82
0.81
Declaration
Health Check-up
Medical Examination
Underwriting Method
Grouped A/E (Traditional Analysis)
PDP

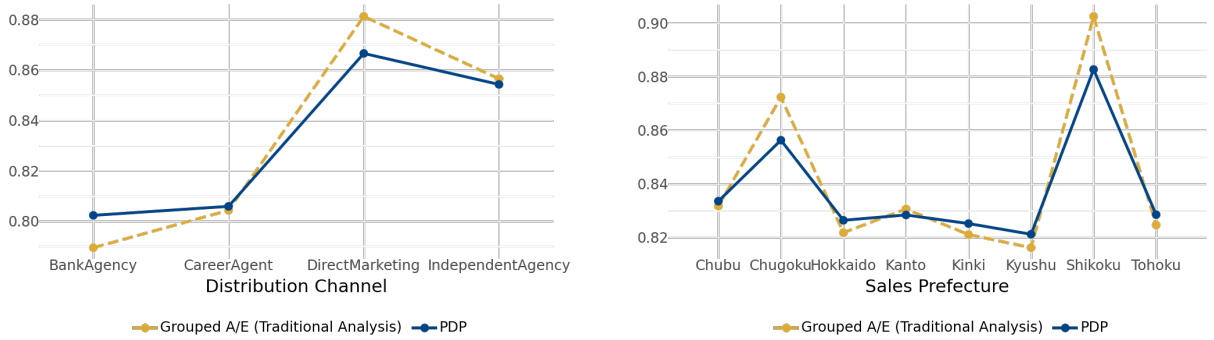


Figure 6: Comparison between observed A/E ratios and PDPs

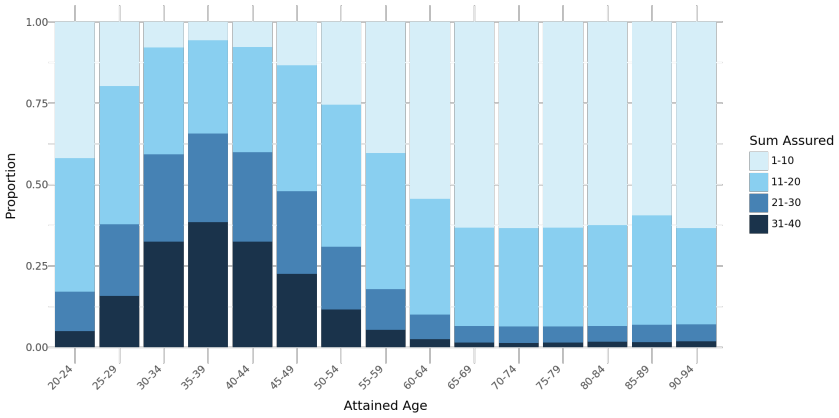


Figure 7: Proportion of Policies by Sum Assured across Attained Age

Attained age was used to compare the results of traditional aggregation-based A/E analysis with those of the PDP. As shown earlier in Figure 1, the aggregation-based analysis indicated relatively high A/E ratios for lives insured aged 30–49. In contrast, the PDP for attained age in Figure 6 did not exhibit a pronounced peak for ages 30–49; instead, the predicted A/E ratio increased gradually with attained age.

The two analyses describe different quantities. The aggregation-based A/E ratio for each age group reflects not only attained age but also the age-specific composition of the other features. By contrast, the PDP evaluates the fitted model at each attained-age value while averaging over a common distribution of the other features. Comparing the two can therefore help analysts explore whether an observed age-specific pattern may partly reflect differences in portfolio composition.

In the simulated data used in this study, the proportion of policies with larger sums assured was relatively high among lives insured aged 30–49, as shown in Figure 7. Since larger sums assured were associated with higher predicted A/E ratios in the fitted model, this difference in portfolio composition may contribute to the relatively high aggregated A/E ratios observed for ages 30–49.

The PDP should not, however, be interpreted as identifying an effect of attained age independently of all other features. In particular, when attained age is strongly associated with other features, the PDP may evaluate combinations of feature values that are uncommon in the observed data, as discussed in Section 2.3.

5.1.4 Interaction Analysis

The results of Friedman's H-statistic are shown in Figure 8.

In traditional A/E analysis, analysts can examine selected combinations of features, but it is not straightforward to identify systematically which feature pairs should be investigated for possible interactions. In this example, Friedman's H-statistic ranked the combination of duration year and sum assured as having the strongest pairwise interaction in the fitted model. Figure 9 therefore presents the corresponding two-way PDP.

The plot suggests that the relationship between sum assured and the predicted A/E ratio varies across policy-duration groups. Higher sums assured are associated with higher predicted A/E ratios within some duration ranges, whereas this pattern is less pronounced in others. Thus, the modeled relationship with sum assured is not uniform across the portfolio but depends on duration year.

This example illustrates how Friedman's H-statistic can be used to screen feature pairs for potentially important interactions and how a two-way PDP can then be used to examine the form of a selected interaction.

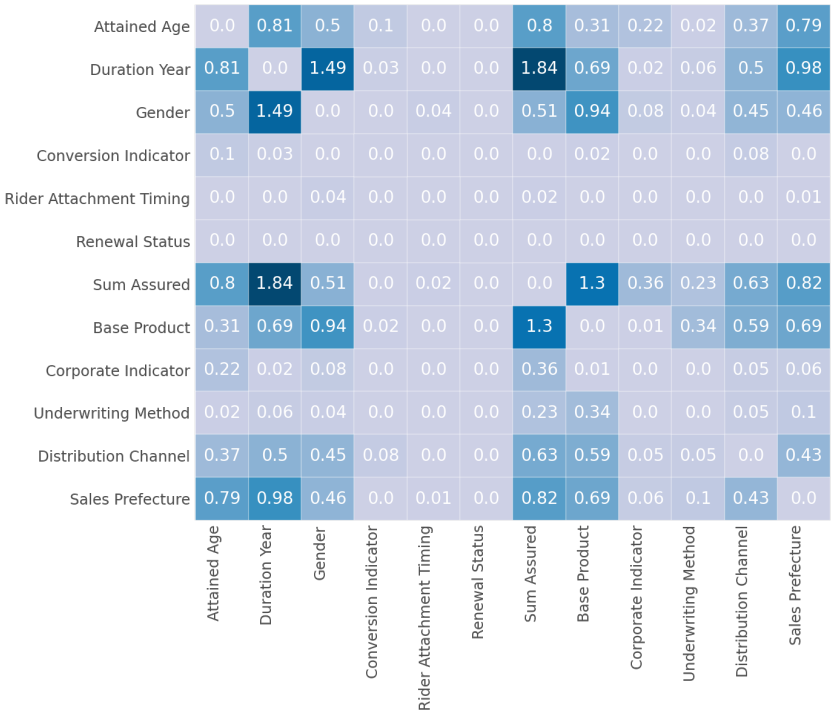


Figure 8: Pairwise interaction strengths measured by Friedman's H-statistic

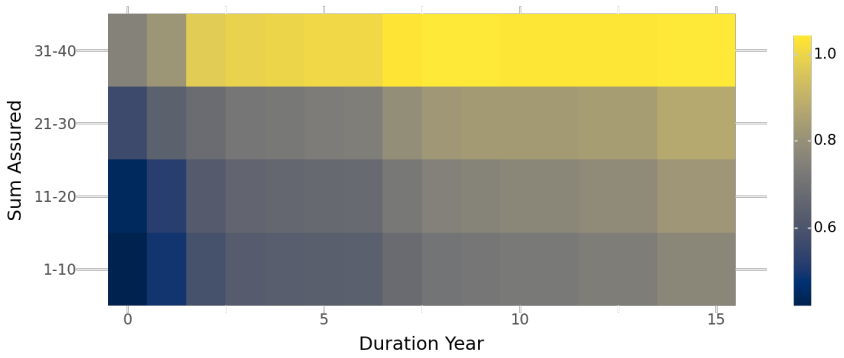


Figure 9: Interaction between duration and sum assured visualized by two-way PDP

5.2 Case 2: Policy-Level Analysis Using SHAP

Figures 11 and 10 present representative SHAP waterfall plots for policies with medium and high predicted A/E ratios, respectively.

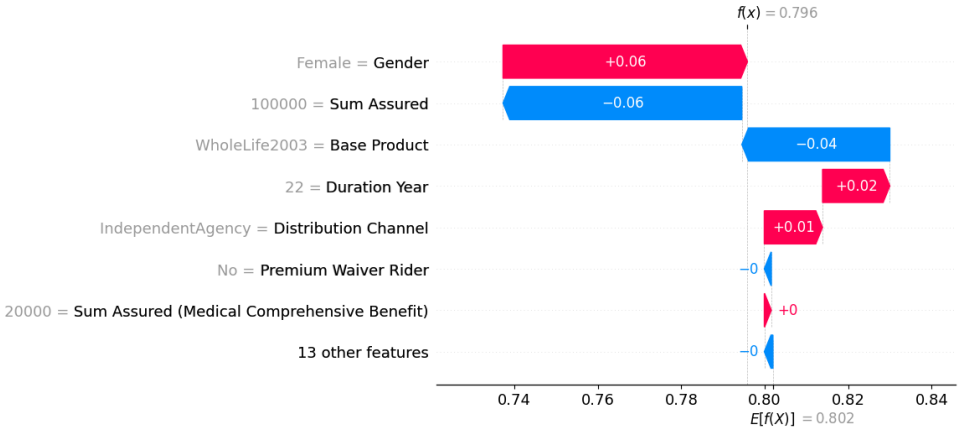


Figure 10: Feature contributions to midium predicted A/E ratio shown by SHAP waterfall plot

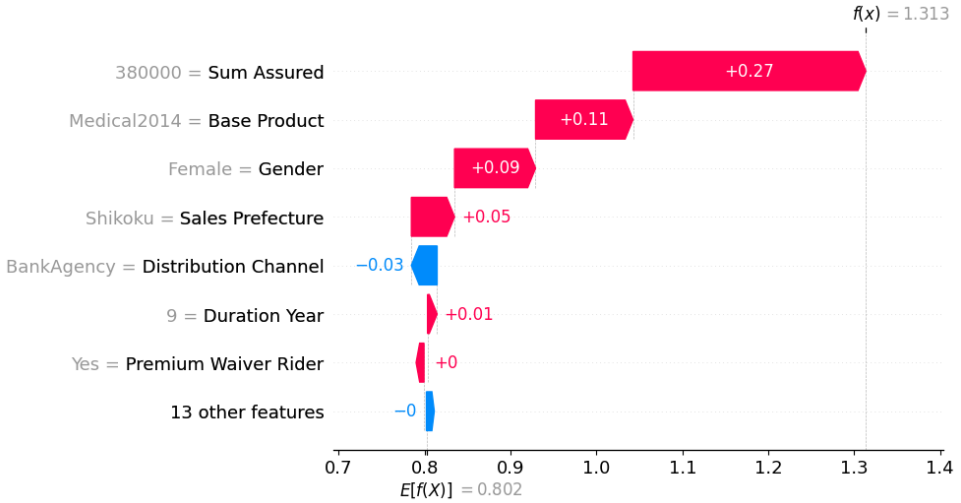


Figure 11: Feature contributions to a high predicted A/E ratio shown by SHAP waterfall plot

Figure 10 shows a policy with a predicted A/E ratio of 0.796, compared with a baseline prediction of 0.802. The main feature contributions included a positive contribution of 0.06 from gender and a negative contribution of 0.06 from sum assured. Overall, the contributions resulted in a predicted A/E ratio close to the baseline.

In contrast, Figure 11 shows a policy with a predicted A/E ratio of 1.313, which is 0.511 higher than the baseline prediction of 0.802. The largest positive contribution came from sum assured (+0.27), followed by base product (+0.11). These contributions substantially increased the predicted A/E ratio.

These examples illustrate how SHAP can be used to examine the feature contributions to the predicted A/E ratio at the individual policy level. By comparing policies with different levels of predicted A/E ratios, analysts can identify which policy characteristics contribute to the differences in predicted A/E ratios and use these findings to guide more targeted investigations and follow-up analyses in actuarial practice.

This section demonstrated how IML can complement the current A/E analysis through the two analytical cases. PFI systematically identified the most influential features, while PDP

revealed the relationships between individual features and the predicted A/E ratio after averaging over the distribution of the other features. Friedman's H-statistic and the two-way PDP identified and visualized interaction effects between features in the fitted model, and SHAP explained the factors contributing to the predicted A/E ratio for individual policies. Together, these results demonstrate that IML provides complementary insights that address the practical limitations of traditional A/E analysis discussed in Section 3.

6 Discussion

6.1 Practical implications for actuarial practice

Traditional A/E analysis remains an essential component of post-pricing validation because it provides a simple, transparent, and well-established framework for monitoring the appropriateness of pricing assumptions. The objective of the proposed framework is therefore not to replace traditional A/E analysis, but to complement it by providing additional insights that are difficult to obtain from aggregation-based analyses alone.

The empirical results demonstrate that IML can provide several types of information that complement traditional A/E analysis. In this study, PFI was used to identify features that were influential in the fitted model, PDP was used to examine the relationships between individual features and model predictions, Friedman's H-statistic and two-way PDP were used to analyze feature interactions, and SHAP was used to examine the contributions of individual features to individual predictions. By combining these analyses, actuaries can investigate the factors associated with high A/E ratios from multiple perspectives and obtain insights that are difficult to derive from traditional aggregation-based analyses.

Another important advantage of the proposed framework is that it can be incorporated into existing actuarial workflows without fundamentally changing current practice. The value of the proposed framework lies not in replacing actuarial expertise, but in enabling actuaries to discover previously unrecognized patterns and generate analytical hypotheses that can be further examined using traditional actuarial methods and professional judgment.

6.2 Limitations

First, this study used simulated policy data designed to mimic the structure of insurance operational data, including the data fields, aggregation framework, and data generation process, while preserving data confidentiality. Although this enables reproducible research, the numerical results should not be interpreted as representing the experience of any particular insurance portfolio.

Second, the interpretations provided by IML depend on the predictive model and data preprocessing procedures, and different models may produce different interpretations. Moreover, IML explains the behavior of the predictive model rather than causal relationships in the underlying data. Consequently, the identified feature effects and interactions should be regarded as exploratory findings. Before being incorporated into actuarial practice, these findings should be evaluated together with actuarial expertise, traditional analytical results, and domain knowledge.

Third, PFI, PDP, and SHAP were selected mainly because they are widely used and are implemented in commonly used software packages. Friedman's H-statistic was used together with two-way PDP to analyze interaction effects. However, the methods used in this study are not the only methods that can be used for these analyses. As discussed in Section 2, each method has limitations. Other methods may be more suitable depending on the purpose of the analysis and the characteristics of the data. For example, Maximum Interpretation Decomposition (MID) is an interpretable machine learning method based on functional decomposition [13]. It can be used to analyze feature importance, main effects, interaction effects, and contributions to

individual predictions within a common framework. Therefore, in practical applications of IML, the methods should be selected according to the purpose of the analysis and the characteristics of the data. The purpose of this study is to demonstrate the usefulness of IML as an EDA tool, rather than to recommend the particular methods used in this study.

6.3 Future work

Although this study focused on post-pricing validation for Japanese private health insurance, the proposed framework is not limited to this application. Many actuarial tasks involve large and complex datasets that require exploratory data analysis to identify important patterns and generate analytical hypotheses. Therefore, the proposed framework has the potential to support a wide range of actuarial analyses.

Future research should demonstrate the applicability of the proposed framework to other areas of actuarial practice, such as mortality analysis, lapse analysis, underwriting analysis, and economic value-based risk management. Accumulating practical case studies across different actuarial applications would contribute to establishing IML as a practical exploratory data analysis tool for actuarial practice.

7 Conclusion

This paper proposed a framework for applying IML as an EDA tool for post-pricing validation in Japanese private health insurance. By applying PFI, PDP, Friedman's H-statistic, two-way PDP, and SHAP, the proposed framework enables actuaries to analyze influential features, feature effects, feature interactions, and policy-level explanations from multiple perspectives.

The proposed framework complements traditional A/E analysis and demonstrates how IML can be incorporated into actuarial practice as a practical EDA tool for understanding complex insurance data and generating new analytical hypotheses. We hope that this study will encourage the wider adoption of IML in actuarial practice and contribute to further development of data-driven actuarial analysis.

Code and data availability

To facilitate reproducibility and further application of the methods presented in this study, the Python code and simulated dataset used in the analysis are publicly available on GitHub at `https://github.com/Kaori-Maruyama/iml-post-pricing-validation`. The simulated dataset is designed to reproduce the key characteristics of operational data while preserving confidentiality. The repository also includes the code used for data generation, model fitting, and the IML analyses presented in this paper.

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Appendix A. Examples of Variable Aggregation Used in the Analysis

In the analytical database used for A/E analysis in actuarial practice, categorical variables such as product type and sales channel often contain a large number of categories. This is because policy information is retained at a detailed level so that the portfolio can be analyzed from various perspectives when necessary, rather than because every category is regularly analyzed.

In practice, categorical values such as product codes are also typically designed according to consistent naming or coding rules. For example, products belonging to the same base product often share common names or code structures, making it straightforward to identify related products and aggregate categories according to the analytical objective.

Following this practical approach, several categorical variables were aggregated into actuarially meaningful groups in this study to facilitate exploratory data analysis using IML. Examples of these aggregations are presented below.

Table 8: Example of Category Aggregation

Code	**Product Name**	**Base Product Category**
01	Term1992	Term
02	Term2001	Term
03	Term2008	Term
04	Term2016	Term
05	Term2022	Term
11	RenewableTerm1998	Renewable Term
12	RenewableTerm2007	Renewable Term
13	RenewableTerm2015	Renewable Term
14	RenewableTerm2021	Renewable Term
21	WholeLife1994	Whole Life
22	WholeLife2003	Whole Life
23	WholeLife2012	Whole Life
24	WholeLife2020	Whole Life
31	Medical2000	Medical
32	Medical2008	Medical
33	Medical2014	Medical
34	Medical2020_A	Medical
35	Medical2020_B	Medical
36	Medical2024_A	Medical
37	Medical2024_B	Medical
41	Cancer1997	Cancer
42	Cancer2009	Cancer
43	Cancer2021	Cancer

Appendix B. Computational Time of Model Training and IML Methods

B.1. Purpose

To assess the practical applicability of IML in actuarial practice, the computational time required for model training and each IML method was measured. Because computational time depends on the execution environment and software implementation, the results presented here should be regarded as illustrative rather than absolute performance benchmarks. Nevertheless, they provide useful reference information for comparing the relative computational costs of the different IML methods and their sensitivity to data size.

B.2. Computational Environment and Analysis Setup

B.2.1. Computational Environment

Table 9 summarizes the computational environment used in this study. All computations were performed using CPU implementations.

Table 9: Computational Environment

Item	**Specification**
CPU	Intel(R) Xeon(R) CPU @ 2.20GHz
Memory	12 GB
Operating System	Ubuntu 22.04.5 LTS
Python	3.12.13
LightGBM	4.6.0
scikit-learn	1.6.1
SHAP	0.52.0

B.2.2. Data and Analysis Setup

The computational times reported in this appendix cover both Dataset A and Dataset B. Table 10 summarizes the data and analysis settings for each IML method.

For each dataset, the underlying data were divided into training and test datasets, and the predictive model was trained using the training data. PFI and the SHAP summary plot were calculated using the test data, whereas PDP, Friedman's H-statistic, and two-way PDP were calculated using the training data. The SHAP waterfall plot was calculated using the test data of Dataset B.

The analysis used 12 explanatory variables. PFI, Friedman's H-statistic, and the SHAP summary plot were calculated using all 12 variables. One-way PDPs were calculated for each of the 12 variables. Friedman's H-statistic was calculated for all 66 pairwise combinations of the 12 variables. A two-way PDP was calculated for one selected pair, duration year and sum assured.

For the H-statistic, 10% of the Dataset A training data were randomly sampled to reduce computational time. The SHAP waterfall analysis used all of the Dataset B training data.

For the data used in Section 5, Dataset A contains 51,400 test records after aggregation, while Dataset B contains 600,000 records before the train-test split. The SHAP summary plot was calculated using all 51,400 test records in Dataset A.

Table 10: Analysis Setup for Each IML Method

IML Method	Scope	Dataset	Data Used	Variables / Combinations
PFI	Global	A	Test	12 variables
PDP	Global	A	Training	12 variables
H-statistic	Global	A	Training[1]	66 pairs
Two-way PDP	Global	A	Training	1 pair
SHAP summary plot	Global	A	Test	12 variables
SHAP waterfall plot	Local	B	Test	Individual policies

[1] A random sample corresponding to 10% of the training data was used to reduce computational time.

B.2.3. Software and Implementation

The IML methods were implemented using the Python packages and functions summarized in Table 11. PFI and PDP were calculated using functions provided by scikit-learn. SHAP values and the corresponding plots were generated using the SHAP package.

Friedman's H-statistic was calculated using custom code based on one-way and two-way partial dependence values. No dedicated Python package was used for the calculation; instead, the statistic was implemented directly following the definition described in Section 2.

Table 11: Python Packages and Functions Used for IML

IML Method	Package	Function / Implementation
PFI	scikit-learn	`permutation_importance()`
PDP	scikit-learn	`partial_dependence()`
H-statistic	Custom implementation	Based on partial dependence values
Two-way PDP	scikit-learn	`partial_dependence()`
SHAP summary plot	SHAP	`TreeExplainer()`, `beeswarm()`
SHAP waterfall plot	SHAP	`TreeExplainer()`, `waterfall()`

B.3. Computational Time by Dataset Size

Table 12 summarizes the computational time required for model training and each IML method for datasets of different sizes. The number of records in the first column denotes the size of the underlying dataset before the train-test split.

The computational times of the IML methods should be interpreted in light of the scope of each analysis. PFI, PDP, Friedman's H-statistic, two-way PDP, and the SHAP summary plot are global analyses, whereas the SHAP waterfall plot provides local explanations for individual policies. In addition, the computational times reflect different numbers of variables or feature combinations, as described in Table 10. Therefore, the computational times are not intended to provide a direct performance ranking across IML methods, but rather to illustrate the computational cost of the analysis settings used in this study.

The results show that all IML methods were completed within approximately 44 minutes, even for the largest dataset with 1,200,000 records. This suggests that the computational time of the IML methods examined in this study is not a major practical constraint in actuarial practice.

Table 12: Computational Time by Dataset Size (seconds)

No. of Records	Model Training	PFI	PDP	H-statistic	Two-way PDP	SHAP summary plot	SHAP waterfall plot
6,000	0.42	14.88	17.42	55.06	13.32	1.57	5.84
60,000	2.07	108.39	117.96	325.48	111.46	7.66	13.5
600,000	12.71	617.46	662.16	1828.28	648.74	40.51	103.85
1,200,000	20.43	903.01	965.24	2614.68	916.9	57.15	196.67

Appendix C. Effect of Dataset Size on IML Results

The computational environment was the same as that described in Appendix B. The pseudo-dataset used in this study was sufficiently large to ensure stable model estimation. In actuarial practice, however, the number of policies and the expected claim amount vary considerably across analytical segments. Consequently, the amount of available data may influence the results obtained from IML methods. This appendix examines how dataset size affects the results of representative IML methods by comparing analyses conducted using datasets of different sizes.

We compare the IML results using datasets with three different sizes: 6,000, 60,000 and 600,000 records. The datasets were generated using the same data generation procedure as the main analysis.

Figures 12–14 compare the results of representative IML methods obtained from datasets with different sizes. As the dataset size increased, the results of the IML methods became more consistent with those obtained from the largest dataset. For PFI, Figure 12 shows that the main influential features, such as duration year, attained age, and sum assured, could still be identified even with the smaller datasets. However, the ranking of feature importance differed from that obtained with the larger datasets. For PDP, Figure 13 shows that smaller datasets made it more difficult to capture the relationships between the A/E ratio and individual features. For example, the relationship between attained age and the predicted A/E ratio was less clearly captured with smaller datasets. For the SHAP waterfall plots, Figure 14 shows that the feature contributions also differed substantially across dataset sizes. In particular, with smaller datasets, features that were influential in the larger datasets, such as sum assured, did not always appear among the largest contributions. This suggests that a certain amount of data is needed for the predictive model to learn the feature effects sufficiently for reliable interpretation using IML.

These results suggest that IML methods can still be applied to relatively small datasets, but the results should be interpreted with greater caution when the available data are limited.

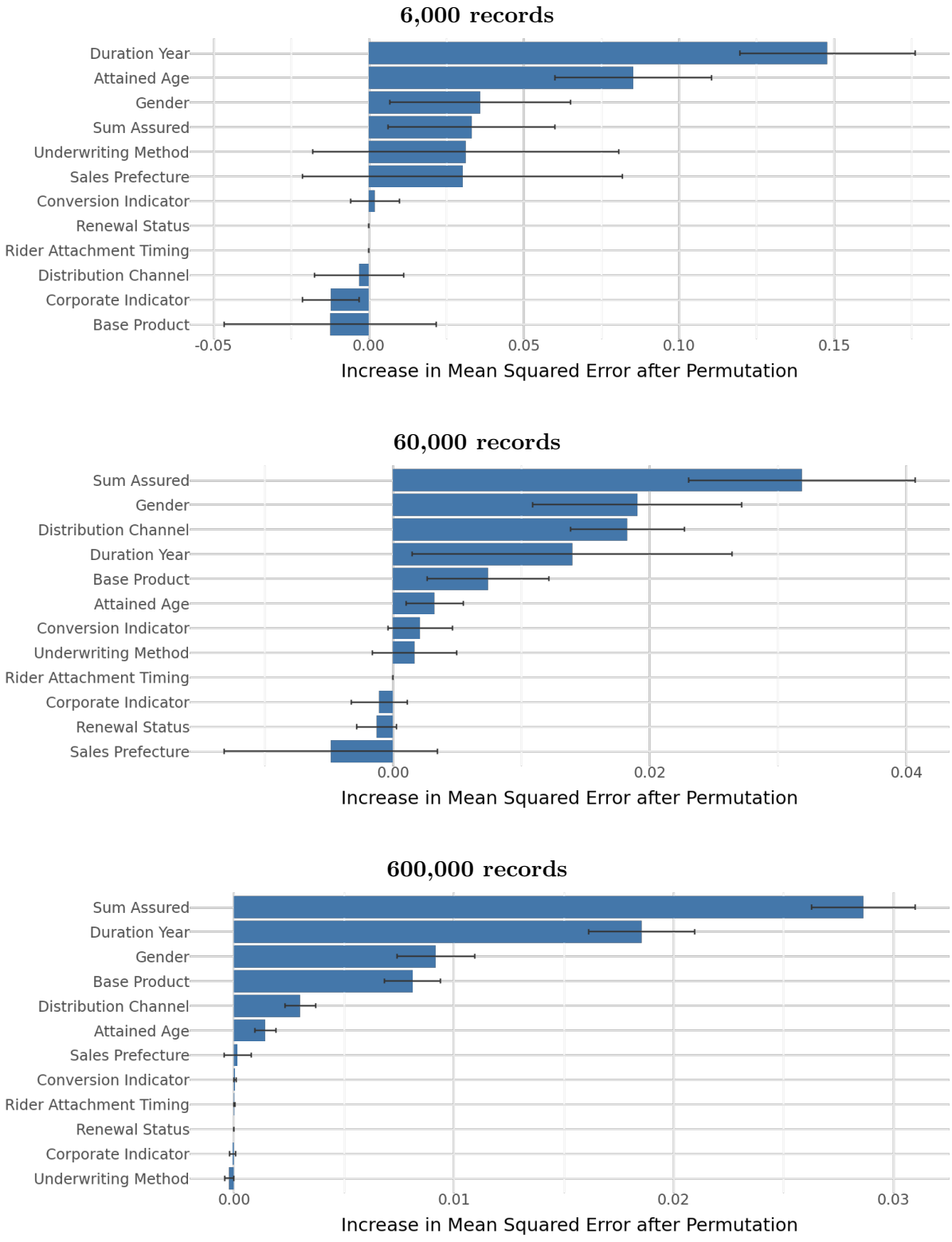


Figure 12: PFI for Different Dataset Sizes

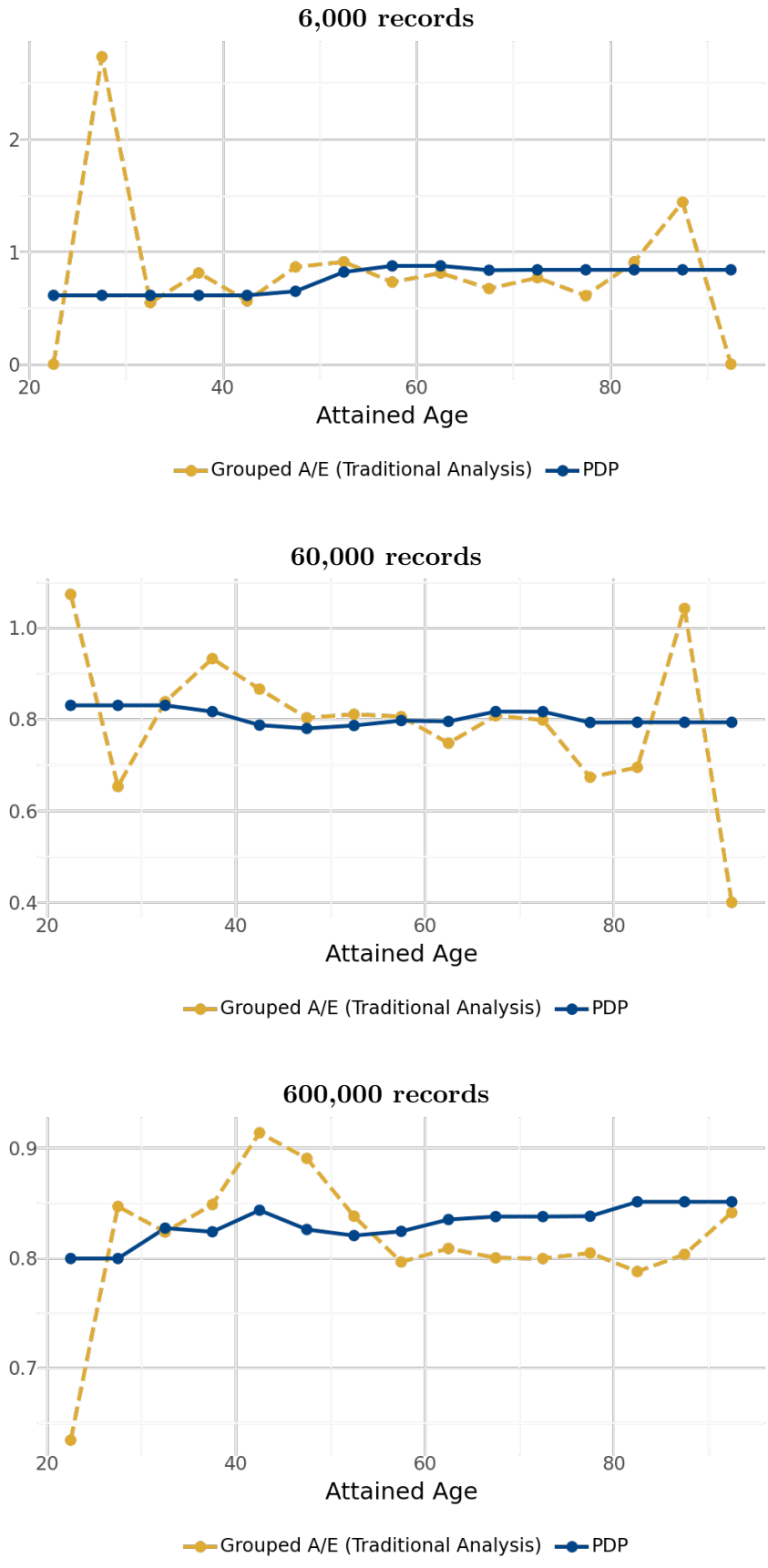


Figure 13: PDP for Different Dataset Sizes

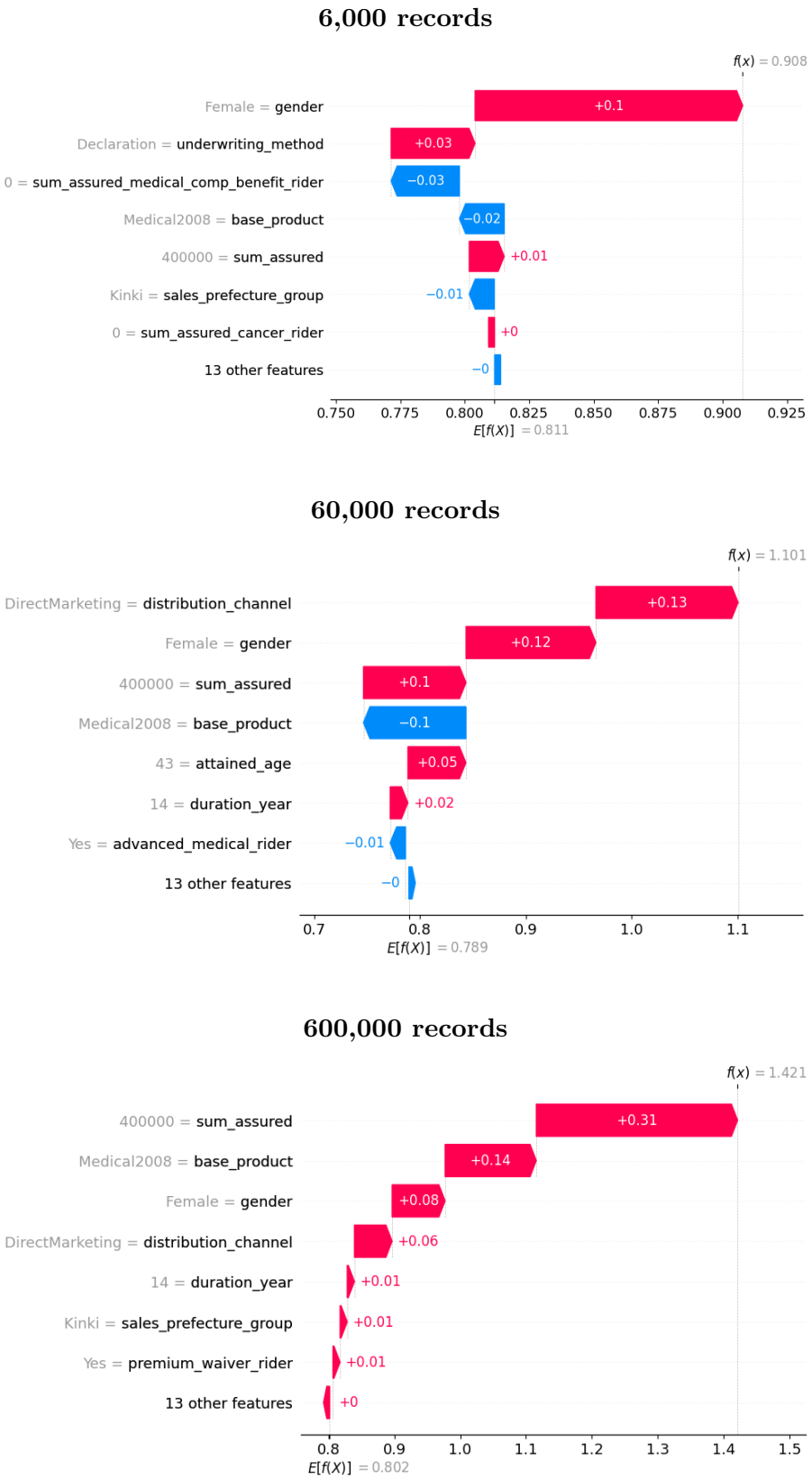


Figure 14: SHAP waterfall plot for Different Dataset Sizes